

Training: mlp (dense network)

Three runs of the two-hidden-layer MLP on the tabular features.

The MLP sees only the engineered features of the newest beat (no RR sequence). It is the cheapest network (about 1,100 MACs per beat) and trains in minutes.

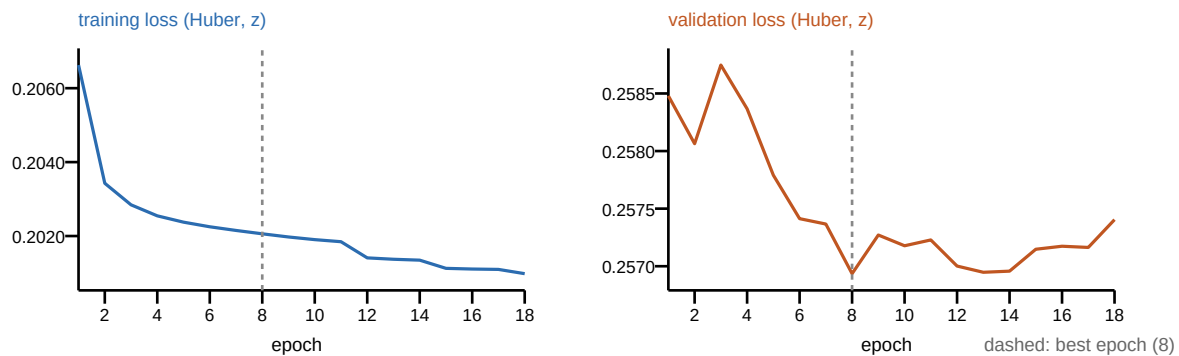
How this family is trained

- Dense(32, ReLU) - Dense(16, ReLU) - Dense(1), Keras default initialisation, residual target, the shared Keras recipe (00_training_pipeline).
- Each run changes only the feature set; architecture, seed and optimiser settings are the same.

Runs

mlp - round 1, 19 features

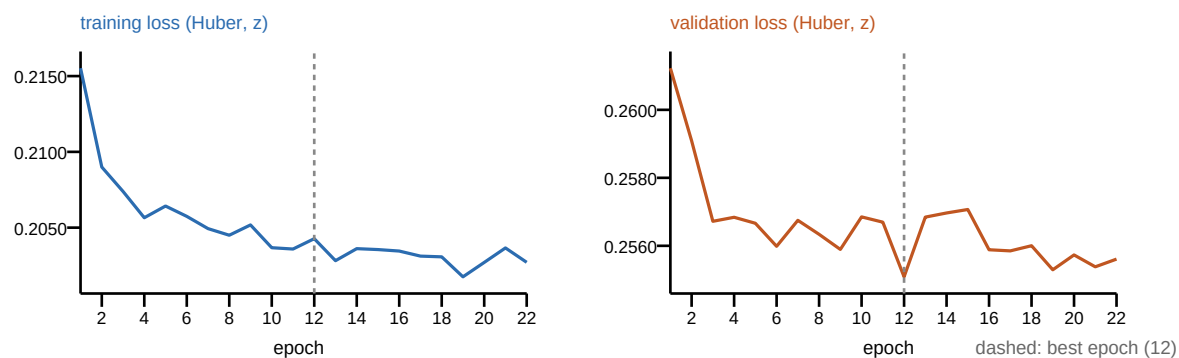
item	value
command	python src/train.py --models mlp --tag mlp
start	from scratch (random initialisation, seed 211)
inputs	19 tabular features (set tab)
target	z-scored correction over the RLS forecast (dRR - rls_pred)
epoch	20,755,456 windows = one full pass over the 20,755,925 training windows (round-1 train.py)
budget	18 of at most 100 epochs (early stopping, patience 10); 373,598,208 windows drawn = 18.0 passes over the training set
learning rate	0.001 at the start, halved 3 times on validation plateaus, 0.00013 at the end
best epoch	8 (validation loss 0.2585 after epoch 1, 0.2569 at best); its weights are the saved model
time	0.98 h on one GPU
int8	exported by the round-1 train.py (its calibration settings were not recorded in config.json); full validation: int8 34.75 vs float 34.60 ms (+0.43%)
validation RMSE (float)	34.60 ms on all 1,421,225 validation windows (-6.9% vs rls_ar 37.14)



- Round-1 epochs were full passes (40,538 steps of 512), so 18 epochs are about 18 passes; the median step took about 4 ms.

mlp_nobp - round 2, without bp and bp_prev

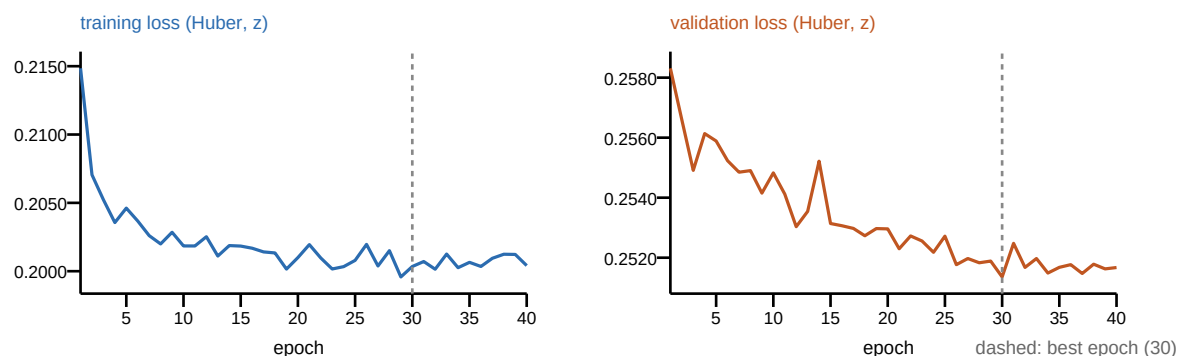
item	value
command	python src/train.py --models mlp --features tab_nobp --tag mlp_nobp
start	from scratch (random initialisation, seed 211)
inputs	17 tabular features (set tab_nobp)
target	z-scored correction over the RLS forecast (dRR - rls_pred)
epoch	1,999,872 windows drawn at random with replacement (512 x 3906 steps)
budget	22 of at most 100 epochs (early stopping, patience 10); 43,997,184 windows drawn = 2.1 passes over the training set
learning rate	0.001 at the start, halved 4 times on validation plateaus, 6.3e-05 at the end
best epoch	12 (validation loss 0.2612 after epoch 1, 0.2551 at best); its weights are the saved model
time	0.07 h on one GPU, 2.9 ms per step
int8	re-exported 2026-10-07 after the 500-window calibration bug, min/max calibration on 20,000 random training windows; first 20k validation windows: 34.90 vs 34.68 ms float; full validation: int8 34.56 vs float 34.45 ms (+0.32%)
validation RMSE (float)	34.45 ms on all 1,421,225 validation windows (-7.2% vs rls_ar 37.14)



- Trained with the 500-window int8 calibration bug; re-exported with 20,000 windows the same evening.

mlp_rsa - round 2, + 6 RSA-tracker features

item	value
command	python src/train.py --models mlp --features tab_rsa --tag mlp_rsa
start	from scratch (random initialisation, seed 211)
inputs	23 tabular features (set tab_rsa)
target	z-scored correction over the RLS forecast (dRR - rls_pred)
epoch	1,999,872 windows drawn at random with replacement (512 x 3906 steps)
budget	40 of at most 100 epochs (early stopping, patience 10); 79,994,880 windows drawn = 3.9 passes over the training set
learning rate	0.001 at the start, halved 6 times on validation plateaus, 1.6e-05 at the end
best epoch	30 (validation loss 0.2583 after epoch 1, 0.2514 at best); its weights are the saved model
time	0.14 h on one GPU, 3.0 ms per step
int8	re-exported 2026-10-07 after the 500-window calibration bug, min/max calibration on 20,000 random training windows; first 20k validation windows: 34.72 vs 34.47 ms float; full validation: int8 34.34 vs float 34.21 ms (+0.39%)
validation RMSE (float)	34.21 ms on all 1,421,225 validation windows (-7.9% vs rls_ar 37.14)



- Same re-export as mlp_nobp. The tracker features were ungated at the time.

What the training showed

- All three converge to within 0.4 ms of each other; the learning curves are flat after a few epochs, so the MLP is limited by its inputs, not by the training budget.
- The closed-loop evaluation (not used in training) rejected all three in favour of the TCNs.

Abbreviations and acronyms used in this document

term	meaning
dRR	beat-to-beat change of the RR interval: $dRR[n+1] = RR[n+1] - RR[n]$ (ms); the quantity predicted
HRV	heart rate variability
int8	8-bit integer: weights and activations quantised to 8-bit integers, as run on the microcontroller; _int8 = the int8 version of a run
MAC	multiply-accumulate operation (one multiplication plus one addition): the unit of compute cost on the microcontroller; MACs/beat = how many the model needs per heartbeat
MACs	multiply-accumulate operations (see MAC)
MLP	multilayer perceptron (a fully connected neural network)
mlp	multilayer perceptron (run name)
ms	milliseconds
nobp	run-name part: no band-pass (the fixed RSA band-pass features bp and bp_prev removed)
ReLU	rectified linear unit, $\max(0, x)$ (an activation function)
RR	RR interval: time between two consecutive heartbeats (R waves of the ECG), in ms
TCN	temporal convolutional network (causal, dilated 1-D convolutions)